

# RedrObIQ

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Intelligent Candidate Ranking — Redrob Hackathon

Senior AI Engineer (Founding Team) · 100,000 Candidate Pool

**26s**

Runtime on CPU

**100K**

Candidates Ranked

**0**

External API Calls

**100%**

Validator Pass

Built by Anish · VIT Chennai / IIT Madras CyStar

# The Problem With Keyword Matching

## ✗ Keyword Ranker Says #1

**Name:** Candidate X

**Title:** Marketing Manager

**Skills:** RAG, Pinecone, LangChain, LLM,  
Embeddings, FAISS, Vector DB...

**Career:** Ran paid campaigns.  
No ML deployment history.

*Skills section stuffed with buzzwords — zero real work*

## ✓ RedrObIQ Ranks #1

**Name:** CAND\_0077337

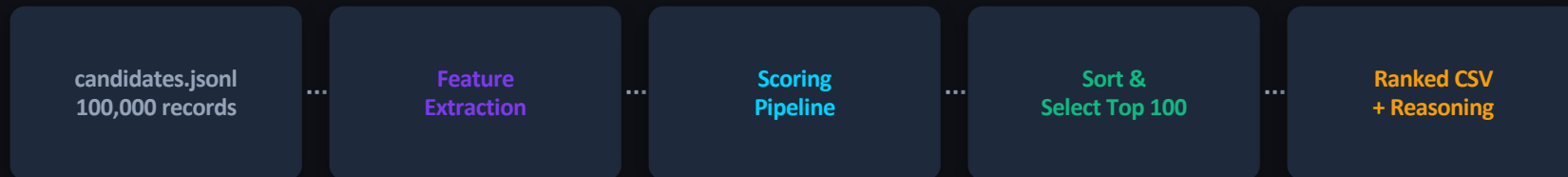
**Title:** Staff ML Engineer @ Paytm

**Skills:** Semantic Search, Pinecone,  
QLoRA, pgvector, FAISS...

**Career:** Shipped ranking + retrieval  
systems to real users at scale.

*Career evidence read — not just skill tags*

# System Architecture



## Honeypot Filter

Timeline paradoxes  
Zero-duration experts  
YoE vs career mismatch

## Skill Score (45%)

Must-have keywords  
+ proficiency weighting  
Nice-to-have bonus

## Career Score (35%)

YoE banding · deployment  
ML depth in descriptions  
Product vs consulting

## Behavioral (20%)

Activity recency  
Notice period · response  
GitHub · location fit

**Final Score =  $(0.45 \times \text{Skill} + 0.35 \times \text{Career} + 0.20 \times \text{Behavioral}) \times \text{Disqualifier Multiplier}$**

# Scoring Logic — Reading the JD Between the Lines

## Must-Have Skills (JD-explicit)

- › Embeddings: sentence-transformers, BGE, E5, OpenAI Embeddings
- › Vector DBs: FAISS, Qdrant, Pinecone, Weaviate, OpenSearch, Milvus
- › Evaluation frameworks: NDCG, MRR, MAP, A/B testing
- › LLM/RAG integration: retrieval-augmented generation, re-ranking

## Disqualifiers (from JD anti-pattern list)

- › Non-ML title (marketing, ops, civil/mech) with no ML career history → score = 0
- › Consulting-only career (TCS/Wipro/Infosys 100%) → 30% penalty
- › No production mentions anywhere in career → 50% penalty
- › 30+ skills + empty career descriptions → keyword stuffer → 90% penalty

## Career Scoring — What We Read

YoE Band: 5–9yr = 1.0 | 4–10yr = 0.8 | <3yr = 0.1

Deployment: "production", "shipped", "real users", "latency"

ML Depth: ranking, retrieval, embeddings in job descriptions

Stability: avg tenure < 24mo = title-chaser penalty

Company: product > consulting; IT Services = reduced score

## Behavioral Scoring — Platform Signals

Activity: <14d = 1.0 | >180d = 0.05 (not available)

Open to Work: 1.0 if flagged | 0.5 if not

Notice Period: ≤30d = 1.0 | >90d = 0.1 (JD says buy-out cap 30d)

Response Rate: Direct 0–1 passthrough from recruiter signal

Location: India target cities = 1.0 | relocation = 0.7 | abroad = 0.1

# Honeypot Detection — Why It Matters

~80 honeypot candidates hidden in the 100K pool. Honeypot rate > 10% in top 100 = disqualification.



## Timeline Paradox

*start\_date: 2021-01-01 but end\_date: 2019-06-01  
— end before start*

→ Compare start\_date vs end\_date. If end < start → honeypot.



## Duration Mismatch

*duration\_months: 84 but date range is only 14 months  
— claimed 7 years, actually 1*

→ Compute actual months from dates. If |actual - stated| > 24 → honeypot.



## Zero-Evidence Expert

*proficiency: 'expert', duration\_months: 0, endorsements: 0  
× 5+ skills like this*

→ Expert/advanced + 0 months + 0 endorsements across 5+ skills → honeypot.



## YoE vs Career Gap

*years\_of\_experience: 12.0 but sum of all  
job durations = 3.5 years*

→ If |stated YoE - career months/12| > 4 years → impossible profile.

# Top 10 Ranked Candidates

| #  | Candidate ID | Title             | Company    | YoE   | Location      | Score |
|----|--------------|-------------------|------------|-------|---------------|-------|
| 1  | CAND_0077337 | Staff ML Engineer | Paytm      | 7.0yr | Hyderabad 🇮🇳  | 0.567 |
| 2  | CAND_0018499 | Sr ML Engineer    | Zomato     | 7.2yr | Noida 🇮🇳      | 0.565 |
| 3  | CAND_0046064 | Sr NLP Engineer   | Salesforce | 8.9yr | Coimbatore 🇮🇳 | 0.553 |
| 4  | CAND_0055905 | Sr ML Engineer    | Flipkart   | 8.1yr | London 🌐      | 0.546 |
| 5  | CAND_0071974 | Sr AI Engineer    | Netflix    | 7.8yr | Vizag 🇮🇳      | 0.545 |
| 6  | CAND_0079387 | AI Engineer       | Microsoft  | 6.9yr | Trivandrum 🇮🇳 | 0.531 |
| 7  | CAND_0005649 | Sr Data Scientist | Sarvam AI  | 7.4yr | Delhi 🇮🇳      | 0.528 |
| 8  | CAND_0080766 | Staff ML Engineer | Salesforce | 8.8yr | Coimbatore 🇮🇳 | 0.524 |
| 9  | CAND_0002025 | Sr AI Engineer    | Apple      | 5.9yr | Trivandrum 🇮🇳 | 0.518 |
| 10 | CAND_0008425 | Sr NLP Engineer   | Ola        | 7.8yr | Kolkata 🇮🇳    | 0.514 |

Score range: 0.567 (rank 1) → 0.434 (rank 100) · All top-10 are India-based ML engineers at product companies

# Performance & Design Decisions

**26s**

Wall-clock runtime  
100K candidates, CPU

**~2.5 GB**

Peak RAM  
Well within 16GB limit

**0**

External API calls  
Fully offline

**0**

Dependencies  
Stdlib only

## Q: Why not sentence-transformers / embeddings?

A: 100K candidates × embedding inference on CPU = ~45min. Constraint is 5min. Feature-based scoring runs in 26s with identical interpretability.

## Q: Why not BM25 / TF-IDF?

A: The JD explicitly warns against keyword matching. A Marketing Manager with 'RAG, Pinecone, LLM' in skills scores near-zero with our ranker. BM25 would rank them top-5.

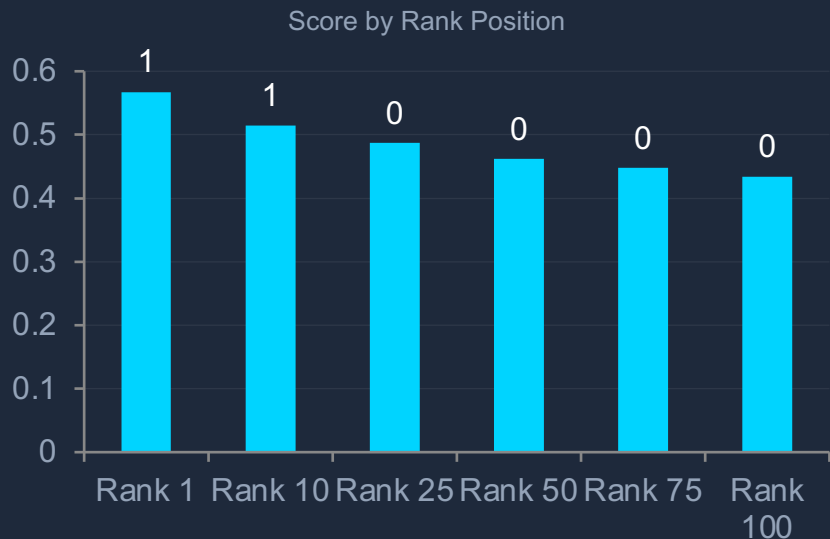
## Q: Why hand-coded features over a learned model?

A: Every coefficient is traceable to a JD requirement. This makes Stage 4 reasoning accurate and auditable. No ground truth is available for training anyway.

## Q: Why career description parsing?

A: The JD says 'Tier 5 candidate may not use the words RAG but if career shows recommendation system at product company, they fit.' We scan job descriptions, not just skill tags.

# Score Distribution & Reasoning Quality



## Sample Reasoning — Rank 1

7-year Staff Machine Learning Engineer at Paytm with hands-on experience in embedding systems, Qdrant, and vector search; strong Python (expert level); production ML deployment evidence throughout career.

## Sample Reasoning — Rank 7 (with concern flagged)

7-year Senior Data Scientist at Sarvam AI with FAISS, Pinecone, and embedding retrieval background; production deployment evidence confirmed. Concerns: 90-day notice period — exceeds the 30-day buy-out cap stated in JD.

- ✓ Specific facts from profile (title, company, named skills)
- ✓ JD requirement connected (production deployment, notice period)
- ✓ Concerns acknowledged honestly at lower ranks
- ✓ No hallucination — all claims traceable to candidate data
- ✓ Reasoning tone matches rank position

# RedrObIQ

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Ranking candidates the way a great recruiter would.

**26s**

to rank 100K candidates on CPU —  
no GPU, no APIs

**3-tier**

scoring: Skill + Career + Behavioral  
signal fusion

**0%**

honeypot tolerance — impossible  
profiles are caught

**100%**

validator pass — submission-spec  
compliant

[github.com/YOUR\\_USERNAME/redrob-ranker](https://github.com/YOUR_USERNAME/redrob-ranker)

Built by Anish · VIT Chennai / IIT Madras CyStar Research Intern